

Golf

Note: Research has shown that golf-specific exercise programs improve strength, flexibility, and balance in golfers. Improvements in these areas result in increased upper-torso axial rotational velocity, which results in increased club-head speed, ball velocity, and driving distance (Lephart et al., 2007).

Better golfers possess unique physical characteristics that are important for greater proficiency. These characteristics have also been demonstrated to be modifiable through golf-specific training programs (Sell et al., 2007).

Implementing a progressive functional training program including flexibility core, balance, and integrated resistance exercises has been shown to provide significant improvements in club-head speed and several components of functional fitness (Thompson et al., 2007). Golf may not look as physically demanding as other sports; however, the entire movement system must be functioning and trained to perform at the highest levels. Correcting muscle imbalances, increasing flexibility, and improving functional performance have been shown to improve a golfer's on-course results. The Sports Performance Coach should consider a progressive and systematic program for golfers looking to prevent overuse injury and improve play.

All programs should be designed based on an assessment that identifies the needs of the athlete, then strategically applies the assessment results to the training plan. The nature of golf, probably more than any other sport, is defined by one repetitive movement (i.e., the golf swing) and requires a blend of mobility and stability to properly execute the skill of the golf swing.

In many places, golf is a year-round activity and is not defined by a season (in-season/off-season). Considering that many athletes play or practice year-round, and there is not a truly defined season for most, golf programming strategies will follow an undulating programming model. Because golf requires a unique blend of stabilization, strength, and power, an undulating model can address these needs within a shorter period. The Sports Performance Coach must understand the functional movement of the golf swing; however, any golf-specific instruction should be left to a trained golf-teaching professional.

Demands of the Sport

- Golf is played by a wide range of age groups.
- Physical demands, regardless of age, are typically the same for each individual.
- This sport consists of repetitive movement patterns in the transverse plane.
- Golf can cause chronic musculoskeletal problems from repetitive movement patterns (Gosheger et al., 2003).
- Cardiorespiratory conditioning may not seem as important as in other sports; however, the average golfer walks between 4 and 5 miles per round, and endurance can become a factor in a golfer's performance.
- A proper golf-specific warm-up prior to play and practice has shown a significant increase in golfers' performance compared with not performing a warm-up (Fradkin et al., 2007). The Sports Performance Coach should consider designing a proper warm-up for a golfer beyond their strength and conditioning program. A proper warm-up that includes dynamic stretching has also been shown to reduce golf injuries and improve performance (Ehlert & Wilson, 2019).
- Golf requires a unique combination of stabilization, strength, and power throughout the entire human movement system and specifically in the spine, hips, and shoulders.
- Muscular imbalances throughout the human movement system, described as overactive muscles (tightness) and underactive muscles (weakness), can cause faulty movement patterns and can limit the golfer's ability to swing effectively.
- Flexibility has been shown to be the most important variable regarding performance and injury through clinical trials.

Suggested Assessments for Golf

Assessment Type	Assessment Method
Transitional assessments	• Overhead squat • Single-leg squat • Split squat • Star excursion balance test
Mobility assessments (if needed)	• Foot and ankle testing • Lumbo-pelvic-hip complex testing • Shoulder and spine testing
Dynamic movement assessments	• Drop jump assessment • Shark skill test • Davies test
Aerobic test: Choose one of the following	• 12-minute run test • 1.5-mile run test • Beep test
Muscular endurance assessments	• Push-up test • Pull-up test • Curl-up test
Muscular strength assessments	• 1–5 RM bench press • 1–5 RM squat
Power assessments	• Rotational medicine ball throw test (left and right) • Power clean assessment

Annual OPT™ Golf Plan (January–December)

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Sample Monthly OPT Golf Schedule

Month 1

Week 1: 2 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Corrective	OFF	OFF	Stabilization Endurance	OFF	Warm-up	OFF
Other						Play	

Week 2: 3 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Strength Endurance	OFF	Corrective	Stabilization Endurance	OFF	Warm-up	OFF
Practice						Play	

Week 3: 2 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	OFF	Corrective	OFF	Power	OFF	Warm-up	Warm-up
Other						Play	Play

Week 4: 3 days/week

	MON	TUE	WED	THU	FRI	SAT	SUN
OPT Phase	Strength Endurance	OFF	Stabilization	OFF	Warm-up	OFF	Corrective
Other						Play	

Corrective Exercise Goals and Considerations for Golf

Goals	Considerations
- Correct impaired movement patterns and muscle imbalances based on movement assessment results - Improve: - Range of motion, joint stability, and muscle strength and endurance - Total-body stabilization - Consistently implement and perform the corrective program, due to the repetitive nature of the golf swing - Incorporate balance training to improve sensorimotor and proprioceptive skills	- Proper warm-up will be created that incorporates the corrective exercise program. - Evaluation of the assessment results and the physical demands of the swing will lead to a further understanding of which muscles must have the proper amount of range of motion throughout the golf swing.

Sample OPT™ Golf Programs

Corrective Exercise Training

INHIBIT				Sets	Reps	Time	
Myofascial Rolling							
Tensor fascia latae				1		30 sec	
Piriformis				1		30 sec	
Lats				1		30 sec	
Thoracic spine				1		30 sec	
LENGTHEN				Sets	Reps	Time	
Gastrocnemius stretch				1		30 sec	
Standing biceps femoris stretch				1		30 sec	
Supine piriformis stretch				1		30 sec	
Floor lat stretch				1		30 sec	
ACTIVATE			Sets	Reps	Tempo	Intensity	Rest
Side-lying hip abduction			2	12	4/2/1		0 sec
Standing hip adduction			2	12	4/2/1		0 sec
Quadruped arm opposite-leg raise			2	12	4/2/1		90 sec
INTEGRATE			Sets	Reps	Tempo	Intensity	Rest
Total body	Cable squat to row: Single-arm		2	12	4/2/1	60%	90 sec

Strength and Conditioning Goals and Considerations for Golf

Goals	Considerations
• Continually correct and maintain impaired movement patterns based on assessment results • Maintain: ○ Physical health (decreased chance of injury for participation in practice/games, and optimal performance within such activities) ○ Range of motion, joint stability, and muscle strength/endurance ○ Cardiorespiratory endurance (aerobic) ○ Total-body stabilization, strength, and power • Incorporate the power phase into the program to improve velocity and rate of force production	• A systematic plan that implements an undulating programming model, integrating corrective exercise, stabilization, strength, and power, will be developed. • Understanding that the golf swing contains elements of stability, strength, and power will be essential. • Total training volume, including sport-specific training, must be considered.

Phase I: Stabilization Endurance Training

WARM-UP			Sets	Reps	Time	
Myofascial rolling						
Tensor fascia latae			1		30 sec	
Piriformis			1		30 sec	
Thoracic spine			1		30 sec	
Static Stretch						
Gastrocnemius stretch			1		30 sec	
Supine piriformis stretch			1		30 sec	
Floor lat stretch			1		30 sec	
Erector spinae stretch			1		30 sec	
Dynamic Stretch		Sets	Reps	Tempo		
Lunge with rotation		1	10	Moderate	0 sec	
MB woodchop		1	10	Moderate	0 sec	
CORE AND BALANCE		Sets	Reps	Tempo	Rest	
Side-iso-ab		2	12	4/2/1	0 sec	
Suspension bridge		2	12	4/2/1	0 sec	
Single-leg balance reach		2	12	4/2/1	60 sec	
PLYOMETRICS		Sets	Reps	Tempo	Rest	
Box jump with stabilization		2	6	Hold landing	60 sec	
RESISTANCE		Sets	Reps	Tempo	Intensity	Rest
Total body	Optional					
Chest	Suspension push-up: Feet in straps	2	15	4/2/1	60%	0 sec
Back	SL bent-over cobra	2	15	4/2/1	60%	0 sec
Shoulders	Single-leg cable lift: Single-arm	2	15	4/2/1	60%	0 sec
Legs	Lunge to balance: Frontal plane	2	15	4/2/1	60%	90 sec
COOL-DOWN						
Myofascial rolling and static stretching						

Phase 2: Strength Endurance Training

WARM-UP		Sets	Reps	Time
Myofascial Rolling				
IT band		1		30 sec
Piriformis		1		30 sec
Lats		1		30 sec
Active Stretch				
Standing adductor stretch		1		30 sec
Supine biceps femoris stretch		1		30 sec
Kneeling hip flexor stretch		1		30 sec
Standing pec stretch		1		30 sec
Dynamic Stretch		Sets	Reps	Tempo
Multiplanar lunge with reach		1	10	Moderately fast
Squat to overhead reach		1	10	Moderately fast
CORE AND BALANCE		Sets	Reps	Tempo
Crunch with rotation		2	12	1/1/1
Cable lift		2	12	1/1/1
Step-up to balance: Transverse plane		2	12	1/1/1
PLYOMETRICS		Sets	Reps	Tempo
Power step-up		2	8	Repeating
RESISTANCE		Sets	Reps	Tempo
Total body				
Chest	Incline DB chest press	2	12	2/0/2
	Cable chest press		8	4/2/1
Back	Seated lat pull-down	2	12	2/0/2
	Single-leg cable row		8	4/2/1
Shoulders	Seated overhead DB press	2	12	2/0/2
	SL scaption		8	4/2/1
Legs	DB squats	2	12	2/0/2
	Side lunge to balance		8	4/2/1
COOL-DOWN				
Myofascial rolling and static stretching				

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WARM-UP			Sets	Reps	Time		
Myofascial Rolling							
Tensor fascia latae			1		30 sec		
Piriformis			1		30 sec		
Lats			1		30 sec		
Dynamic Warm-up (Some athletes may still require static stretching prior to dynamic warm-up.)							
Lateral tube walking			1	10	0 sec		
Lunge with rotation			1	10	0 sec		
Russian twist			1	10	60 sec		
CORE AND BALANCE			Sets	Reps	Tempo	Rest	
Cable rotation			2	12	Fast	0 sec	
Cable lift			2	12	Fast	0 sec	
Multiplanar hop with stabilization			2	12	Controlled	60 sec	
RESISTANCE			Sets	Reps	Tempo	Intensity	Rest
Total body	Optional						
Chest	Incline DB press	3	5	1/0/1	85%	0 sec	
	Rotational chest pass		10	Explosive	2% of BW	2 min	
Back	Seated cable row	3	5	1/0/1	85%	0 sec	
	Woodchop throw		10	Explosive	2% of BW	2 min	
Shoulders	Seated overhead DB press	3	5	1/0/1	85%	0 sec	
	Oblique MB throw		10	Explosive	2% of BW	2 min	
Legs	DB squat	3	5	1/0/1	85%	0 sec	
	Squat jump		10	Explosive	BW	2 min	
COOL-DOWN							
Myofascial rolling and static stretching							

Pre-Practice/Play Warm-Up Goals and Considerations for Golf

Goals	Considerations
• Ensure: ○ Warm-up is multiplanar; it increases tissue temperature and elasticity, and activates muscles specific to the golf swing ○ Warm-up transitions from exercises directly into hitting golf balls, and should be implemented as part of the pre-practice or pre-play routine for each session	• Corrective exercise programming and dynamic warm-up activities for a proper warm-up will be integrated.

Warm-Up at Course (Pre-Practice/Pre-Play)

STATIC STRETCH	Sets	Reps	Time
Gastrocnemius stretch	1		30 sec
Floor lat stretch	1		30 sec
Standing biceps femoris stretch	1		30 sec
Standing pec stretch	1		30 sec
Standing hip flexor stretch	1		30 sec
Erector spinae stretch	1		30 sec
DYNAMIC WARM-UP			
MB rotations	1	12	0 sec
Lunge with rotation	1	12	0 sec
MB flexion/extension (use golf club instead of MB)	1	12	0 sec
Single-leg Romanian deadlift	1	12	60 sec